

Practice Quiz 02B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 02

Expressing a connective using negation and implication.

1. Construct a formula equivalent to $P \wedge Q$ using only P, Q and the connectives $\neg$ and $\rightarrow$. Verify your formula with a complete four-row truth table.

Solution One possible formula is $\neg(P \rightarrow \neg Q)$.

P	Q	$\neg Q$	$P \rightarrow \neg Q$	$\neg(P \rightarrow \neg Q)$	$P \wedge Q$
F	F	T	T	F	F
F	T	F	T	F	F
T	F	T	T	F	F
T	T	F	F	T	T

The last two columns agree in every row, so the formulas are logically equivalent. In particular, the conditional $P \rightarrow \neg Q$ is false exactly when P and Q are both true.

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